

Lizbeth Alexandra Torres Flores

Diplôme d'ingénieur généraliste (in progress) · Electronic Engineering

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Electronic Engineering graduate from UNI, Peru, currently pursuing the Diplôme d'ingénieur généraliste at IMT Atlantique through a dual-degree programme. My work spans VIS-SWIR computer vision research, industrial electrical engineering, embedded systems, IoT and Python software.

Technical focus: Python, C/C++, VHDL; ESP32, STM32, EFR32MG24 and FPGA; OpenCV and PyTorch; IoT communications; DIALux and UPS engineering.

Education

IMT Atlantique AUG 2025-PRESENT
Diplôme d'ingénieur généraliste (Bac+5) · Double degree Brest / Rennes, France
Dual degree with UNI. Currently specialising in Embedded and Heterogeneous Systems; prior work in IoT and 5G for Industry 4.0.

Universidad Nacional de Ingeniería (UNI) 2020-2025
Electronic Engineering Lima, Peru
Electronic engineering foundations in circuits, programming, digital systems and control, applied through hardware and software projects.

Professional and research experience

Institut Pascal APR-AUG 2026
Computer Vision Research Intern Université Clermont Auvergne · Clermont-Ferrand, France

- Co-designed and implemented the RASMD benchmark protocol: four weather conditions, both registration directions, 100 image pairs per task and four complementary metrics.
- Developed and evaluated RoMa-MIND, combining dense RoMa matches with MIND-based multimodal refinement.

Result: Evaluated eight weather/direction tasks. RoMa-MIND reached 3.43 px reprojection error versus 14.81 px for ORB/RANSAC on sunny SWIR-to-VIS pairs; no method led across every condition.

Tools: Python · OpenCV · PyTorch · SuperPoint · LightGlue · MAGSAC++ · RoMa · MIND · Makefile · Streamlit

ABB Peru DEC 2024-AUG 2025
Engineering intern · Process Industries / Process Automation Process Industries Division · Lima, Peru

- Modelled electrical rooms in 3D in DIALux and selected luminaire types and placement for indoor, outdoor and emergency lighting.
- Sized UPS batteries and chargers using IEEE 1115 and IEEE 485. Migrated the first part of a previously manual Excel process to Python.
- The Python tool takes input data and generates calculations, plots and an Excel file, avoiding repeated manual work in that stage.

Result: Delivered room-specific DIALux studies; the Python UPS workflow generated calculations, charts and Excel output from input data.

Tools: DIALux · 3D modelling · Python · Excel · Autodesk Navisworks · IEEE 1115 / IEEE 485 · UPS

University of Alberta FEB-MAY 2024
International Research Intern Edmonton, Canada

- Developed parameterized 2D and 3D figures in Python and Jupyter; changing dimensions or inputs automatically generates the corresponding exam images.
- Migrated existing MATLAB analysis code for an urban solar-adoption study to Python.
- Implemented only the first part of the migrated tool in MecSimCalc for interactive use.

Result: Automated examination figures; migrated the solar analysis to Python and implemented its first stage in MecSimCalc.

Tools: Python · Jupyter · MATLAB · 2D/3D visualization · MecSimCalc

MMJ Smart Electronics DEC 2023-NOV 2024
Embedded Systems Intern Lima, Peru

- Worked on several client prototypes using ESP32, STM32, sensors, GPS and communications, from assembly and integration to functional checks.
- Fabricated, soldered and tested PCBs and wrote manuals for boards, equipment and prototypes.
- Tested a locally developed PLC prototype and contributed to its user guide; also worked on measurement devices and electrical cabinets.

Result: Supported client prototypes, PCB/PLC tests and documentation, including electronics and IoT workshops within the internship.

Tools: ESP32 · STM32 · PCB · soldering · GPS · sensors · PLC · electrical cabinets · IoT

Smart City Lab, UNI APR 2023-FEB 2024
Research Intern Lima, Peru

- Developed the complete documented vision and control chain of a line-following robot with Raspberry Pi and OpenCV and tested it on a physical course.
- Designed and simulated a 16-servo biped in SolidWorks Motion and programmed its gait.
- Contributed to a 3D-printed EMG-controlled transradial prosthesis using MATLAB and Python for signal processing.

Result: Demonstrated three physical prototypes in the lab; presented the prosthesis at an engineering-for-health congress.

Tools: Raspberry Pi · OpenCV · Python · SolidWorks Motion · Arduino · EMG · MATLAB · 3D printing

Engineering projects

CEREBRO · sensor-network security SECURE IOT · OCT 2026-ONGOING

- After common Contiki-NG and Cooja onboarding, I will work with a teammate on nRF52840 boards: build and flash, establish initial unsecured communication, then study secured tests and available cryptographic acceleration.
- Result: The project is in progress; there are no final measured metrics or conclusions. Simulation and the comparison interface were initially assigned to the other pair.

Eco4Impact · shared TCO core INDUSTRIAL SOFTWARE · SEP 2025-MAR 2026

- Vectura System needed to compare transport choices by initial investment, operating costs and end-of-life resale value. Eco4Impact and EcoBoatTwin needed common functions without duplicating the model.
- I structured truck and country information in JSON, defined calculation inputs and worked on residual-value formulas, coordinated with the team's common CAPEX, OPEX and RV architecture.
- Result: At completion, the two teams had a shared TCO calculation core with modules and tests for trucks and boats. My individual contribution is distinguished from the overall team output.

Voice commands on EFR32MG24 EMBEDDED AI · 2026

- I built a voice-command prototype on the EFR32MG24 to distinguish ‘open', ‘close' and ‘lights-on' from noise and other speech, then turn a recognized command into a physical LED response.
- I prepared the 16 kHz audio dataset and MFCC pipeline in Edge Impulse, trained and evaluated the classifier, then integrated the model and audio processing in C/C++ with TensorFlow Lite Micro and Micrium OS. I connected the predictions to GPIO and tested the application on the board.
- Result: The public Edge Impulse project reports 95.2% validation accuracy and 92.6% test accuracy for the float32 model. For the EFR32MG24 int8 target, Edge Impulse estimates 183 ms latency, 15.4 KB peak RAM and 31.2 KB flash. A board demonstration confirmed voice-triggered LED control.

8-bit VHDL processor on FPGA DIGITAL DESIGN · 2026

- The IMT Atlantique lab supplied the architecture and some starter files. I completed and extended this educational CPU and documented how to check it.
- I completed the control unit and FSMs, wrote a readable RTL datapath alternative to the supplied netlist, added programs and testbenches, and organized Vivado variants and Nexys A7 documentation.
- Result: The repository has VHDL sources, scripts and reproducible tests. Board photographs document the lab setup; the full automated suite has not yet been successfully rerun and is documented separately.

FPGA traffic-light controller DIGITAL CONTROL · 2026

- A road junction needs a safe light sequence with defined timing. This project realizes the controller as observable digital logic on FPGA.
- I designed the VHDL state machine and timing counters, checked transitions in simulation, then deployed it on Nexys4 and wired LED outputs to an MDF junction model.
- Result: Waveforms and the physical demonstration showed the light sequence.

Portable vital-sign monitor IOT PROTOTYPE · 2024-2025

- My UNI final project explored remote acquisition and transmission of physiological readings as an IoT engineering prototype, without clinical validation.
- I integrated MAX30102, MLX90614 and MPX5050DP sensors on ESP32, programmed acquisition and transmission over LoRaWAN and GSM/SIM800L, and built a Ubidots dashboard.
- Result: Tests showed readings arriving and displaying in real time. The technical contribution is sensor, connectivity and dashboard integration.

Autonomous line-following robot EMBEDDED VISION · 2023

- The vehicle needed to detect the line in its own camera frames and correct its direction as it moved.
- I developed the complete vision and control chain: HSV color segmentation, region of interest, line detection and grouping, lateral offset and steering calculation, and motor actuation from Raspberry Pi.
- Result: The robot followed a controlled test course. The report includes images of detection and the vehicle on the route.

Graph-colouring timetable generator MATHEMATICS & SOFTWARE · 2020-II

- At UNI, incompatible classes or lab sections forced students to rebuild schedules. Some prepared 10-12 options manually.
- I represented courses and sections as vertices and timing or selection conflicts as edges. Graph colouring and search generated compatible combinations and exported alternative schedules.
- Result: The project won second place in the UNI Faculty of Sciences 2020-II mathematics-project competition. The report notes that the search can grow exponentially.

Least-squares fitting and applications APPLIED MATHEMATICS · 2020-I

- Observations seldom lie exactly on a theoretical curve. The project studied parameter selection that minimizes residuals and the limits of subsequent predictions.
- I studied linear fitting through normal equations and nonlinear fitting with Gauss-Newton; the team implemented MATLAB examples and examined a logistic model on COVID-19 data.
- Result: Presented at the UNI Faculty of Sciences 2020-I mathematics-project fair. The data application illustrates a mathematical method, not a validated epidemiological forecast.

Technical skills

Programming and technical software	Python for automation, analysis, data models and interactive tools; C/C++ for embedded systems; MATLAB for migration and signal processing; VHDL for digital design. Git, Linux, Jupyter, Makefile and JSON. **Applied in:** *Institut Pascal · ABB · Alberta · Eco4Impact · CPU*

Embedded and real-time systems	Integration and tests on ESP32, STM32, EFR32MG24, Raspberry Pi and Arduino; GPIO, sensors, audio acquisition and Micrium OS. Zynq-7000, Vitis and embedded Linux are current lab work; Zephyr is introductory knowledge. **Applied in:** *MMJ · voice control · IoT monitor · IMT labs*
FPGA and digital design	VHDL, state machines, counters, simulation, synthesis, bitstreams and tests on Nexys A7/Nexys4 with Vivado. Lab starter modules are distinguished from my CPU extensions. **Applied in:** *8-bit CPU · traffic-light controller*
Computer vision and multimodal registration	OpenCV and PyTorch; image processing, matching, homographies and metric-based comparison. SuperPoint, LightGlue, MAGSAC++, RoMa and MIND in the VIS-SWIR benchmark. **Applied in:** *Institut Pascal · line-following robot*
On-device applied AI	MFCC feature preparation with Edge Impulse and TensorFlow Lite Micro integration for local inference under Micrium OS, validated on EFR32MG24. **Applied in:** *Voice-controlled LED*
IoT, sensors and communication	Sensor and GPS integration, LoRaWAN, GSM/SIM800L, MQTT and Ubidots. Introductory I2C, SPI, Modbus and RS485; Contiki-NG/nRF52840 security is ongoing work. **Applied in:** *IoT monitor · MMJ · CEREBRO*
Electrical engineering and automation	DIALux electrical-room lighting, IEEE 1115/485 UPS sizing, Excel-to-Python automation and Navisworks cable/material lists; PLC and PCB tests. **Applied in:** *ABB Peru · MMJ*

Leadership and presentations

IEEE EMBS UNI Student Chapter 2023

HR director and deputy events director from June to December; volunteer in technical projects throughout the year. Coordinated more than six technical activities and gave an EEG/EMG presentation.

UNI student representation 2022-2024

Elected student representative on the Faculty Council (2022-23); served on academic, research, accreditation, quality and enrolment committees, as well as faculty hiring and promotion committees. Joined the Electronic Engineering Department Council in 2024.

IEEE WIE UNI · Girls in ICT Day 2024

Speaker encouraging girls and young women to consider careers in information and communications technology.

Airbus Fly Your Ideas · Team CÓNDOR 2026

Qualified for round two of the Airbus Fly Your Ideas Challenge with team CÓNDOR.

Selected additional training

SENATI · industrial electronics: Power Electronics (18 h), Basic Industrial Instrumentation (18 h), Industrial Sensors and Actuators (16 h), and Fundamentals of Refrigeration and Air Conditioning (12 h).

MMJ & I&T Bionics: PCB Design and Fabrication, 24 hours (2024).

Languages

Spanish: native · French: B2 · English: B2